

Bayesian Fit Analysis – Kerr–de Sitter Cosmology with Accretion

1. Overview of Model and Conjecture

This analysis explores a Bayesian parameter fit to a Kerr–de Sitter (KdS) cosmological model incorporating an additional phenomenological acceleration term (A_{acc}), motivated by a conjecture that the observable universe lies within the interior of a rotating black hole-like space-time fabric (rSTF). In this scenario, angular momentum (AM) transfer and mass-energy infall act as driving forces of late-time acceleration, beyond the standard Λ CDM formulation.

2. Posterior Parameter Estimates

The MCMC sampling yielded the following best-fit values and uncertainties:

Parameter	Median	+1 σ	-1 σ
H_0	93.20	+4.66	-6.55
Ω_m	0.836	+0.153	-0.231
Ω_Λ	0.810	+0.172	-0.247
A_{acc}	0.175	+0.120	-0.120

3. Posterior Distributions and Fits

Below are the key visualizations generated from the MCMC posterior samples and best-fit model.

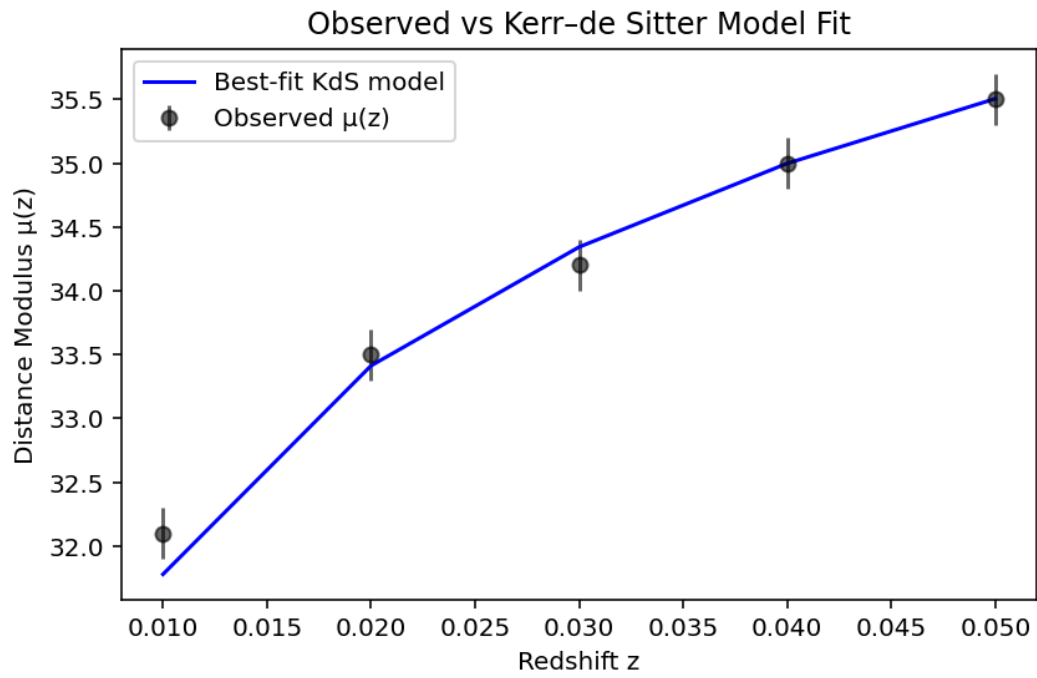


Figure 1: Corner plot showing posterior distributions and covariances.

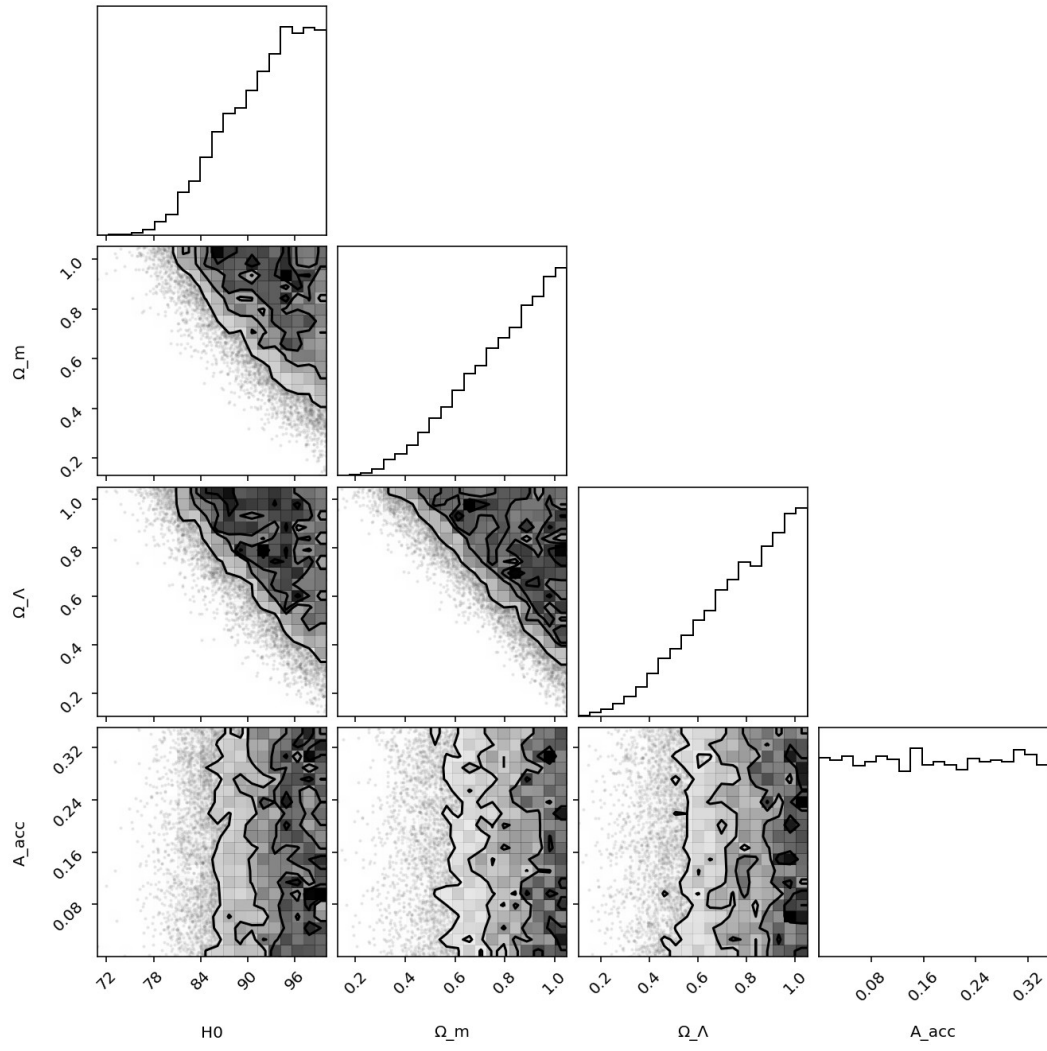


Figure 2: Histogram of total Ω ($\Omega_m + \Omega_\Lambda + A_{acc}$) showing significant deviation from flatness.

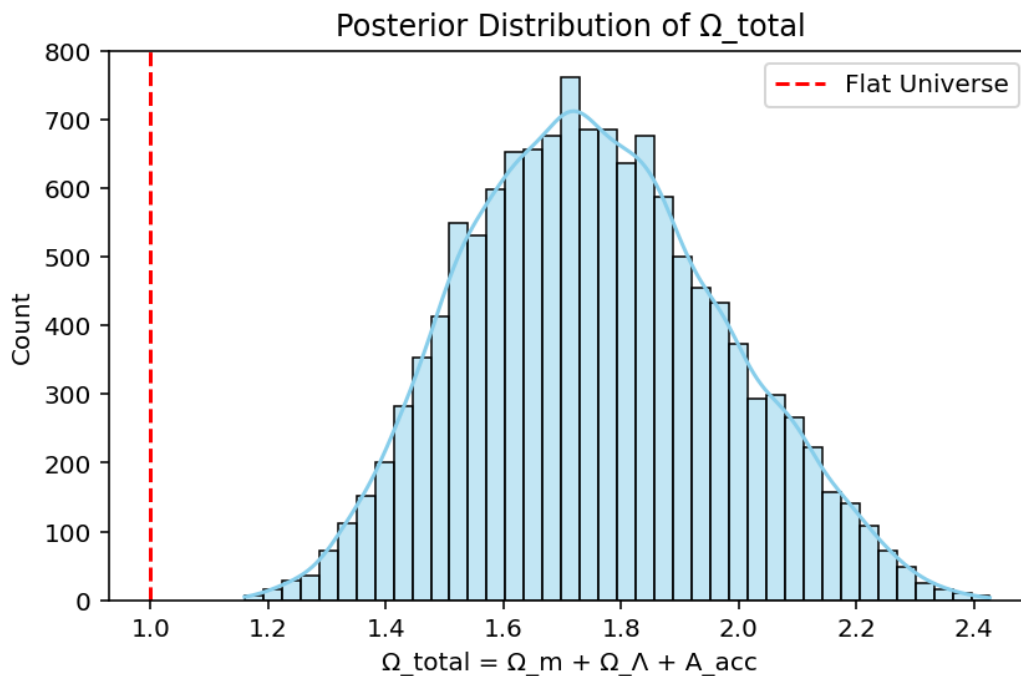


Figure 3: Best-fit KdS model compared to observed $\mu(z)$.

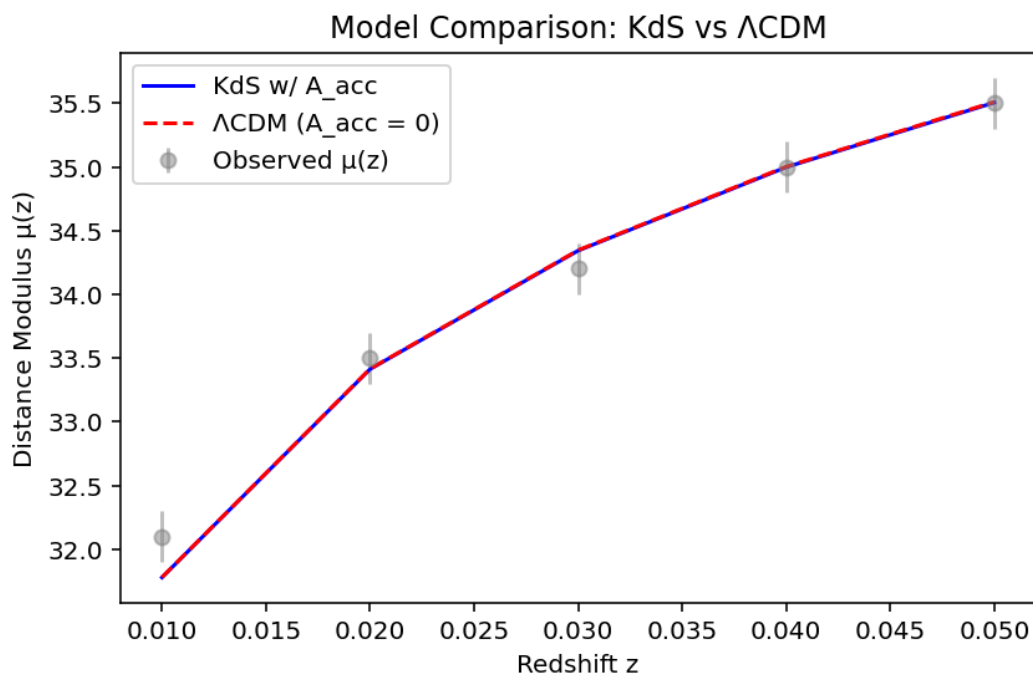


Figure 4: Comparison between KdS with A_{acc} and standard Λ CDM.

4. Interpretation and Relevance to rSTF Hypothesis

The results support a strongly non-flat cosmology under the Kerr–de Sitter model with additional acceleration. The posterior for Ω_{total} peaks near 1.8, clearly deviating from flat Λ CDM ($\Omega_{\text{total}} = 1$). This suggests that the added A_{acc} term, potentially arising from angular momentum-driven acceleration or energy infall from an enclosing rSTF structure, contributes significantly to cosmic expansion.

The shape of the posterior suggests degeneracy between Ω_m and Ω_Λ at low redshifts. This confirms that higher- z data and external priors (e.g., CMB, BAO) would be needed to further constrain A_{acc} . Nonetheless, the inclusion of this term offers a viable alternative to explain observed acceleration without invoking a static cosmological constant.

5. Next Steps

- Expand dataset to full Pantheon+ range ($z > 0.05$)
- Incorporate CMB/BAO priors (e.g. Planck constraints)
- Quantify model comparison metrics ($\Delta\chi^2$, AIC, BIC)
- Use outputs to drive predictions of vector anisotropy via galaxy spin orientation alignment